A SURJECTIVE FLAT MORPHISM OF SCHEMES IS AN EPIMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays the two local ring homomorphisms whose equality is at issue; the display lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 1 navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A flat local homomorphism of local rings is faithfully flat, hence injective. Algebra, Lemma algebra-lemma-local-flat-ff, invoked at proof lines 8717-8719. Allows cancellation of the induced map on local rings, giving equality of the two maps of local rings.
- Equality of morphisms of schemes can be checked on underlying spaces and on local rings. Standard, used at proof lines 8708-8713 and in the conclusion at proof lines 8719-8720. Reduces the assertion that two morphisms are equal to the topological part and the stalkwise part.
- Surjectivity gives that the underlying continuous maps agree. Standard, used at proof lines 8708-8709. Supplies the topological half of the equality.

Source reference. Original lemma 00HR: sources/stacks-chow/tags.txt, source lines 653–653. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *A surjective and flat morphism is an epimorphism in the category of schemes.*

Proof. Suppose we have $h : X' \to X$ surjective and flat and $a, b : X \to Y$ morphisms such that $a \circ h = b \circ h$. As h is surjective we see that a and b agree on underlying topological spaces. Pick $x' \in X'$ and set $x = h(x')$ and $y = a(x) = b(x)$. Consider the local ring maps

$$a_x^\sharp, b_x^\sharp : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$$

These become equal when composed with the flat local homomorphism $h_{x'}^\sharp : \mathcal{O}_{X,x} \to \mathcal{O}_{X',x'}$. Since a flat local homomorphism is faithfully flat (Algebra, Lemma 00HR) we conclude that $h_{x'}^\sharp$ is injective. Hence $a_x^\sharp = b_x^\sharp$ which implies $a = b$ as desired. □

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